Database Systems

Final Project

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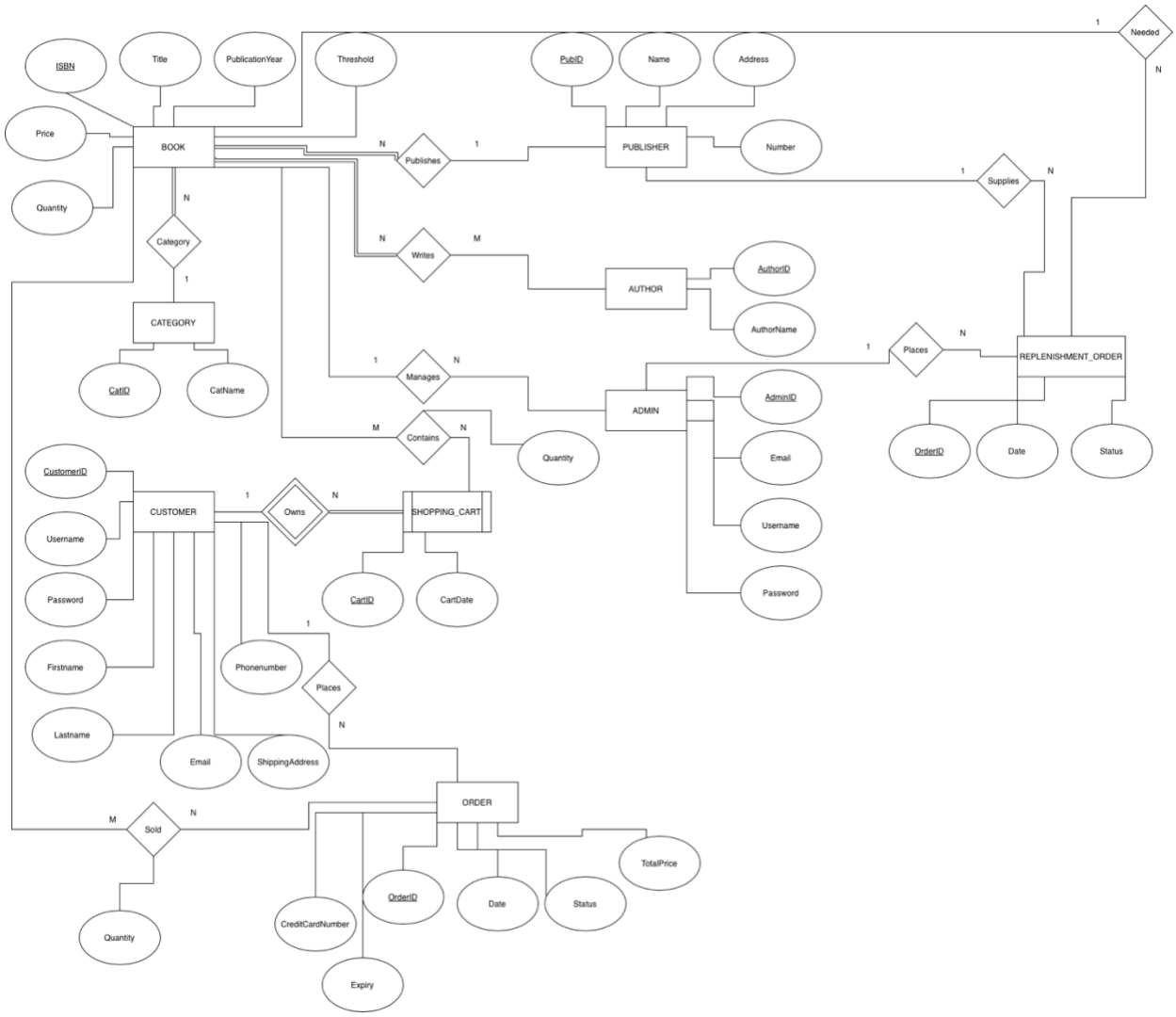
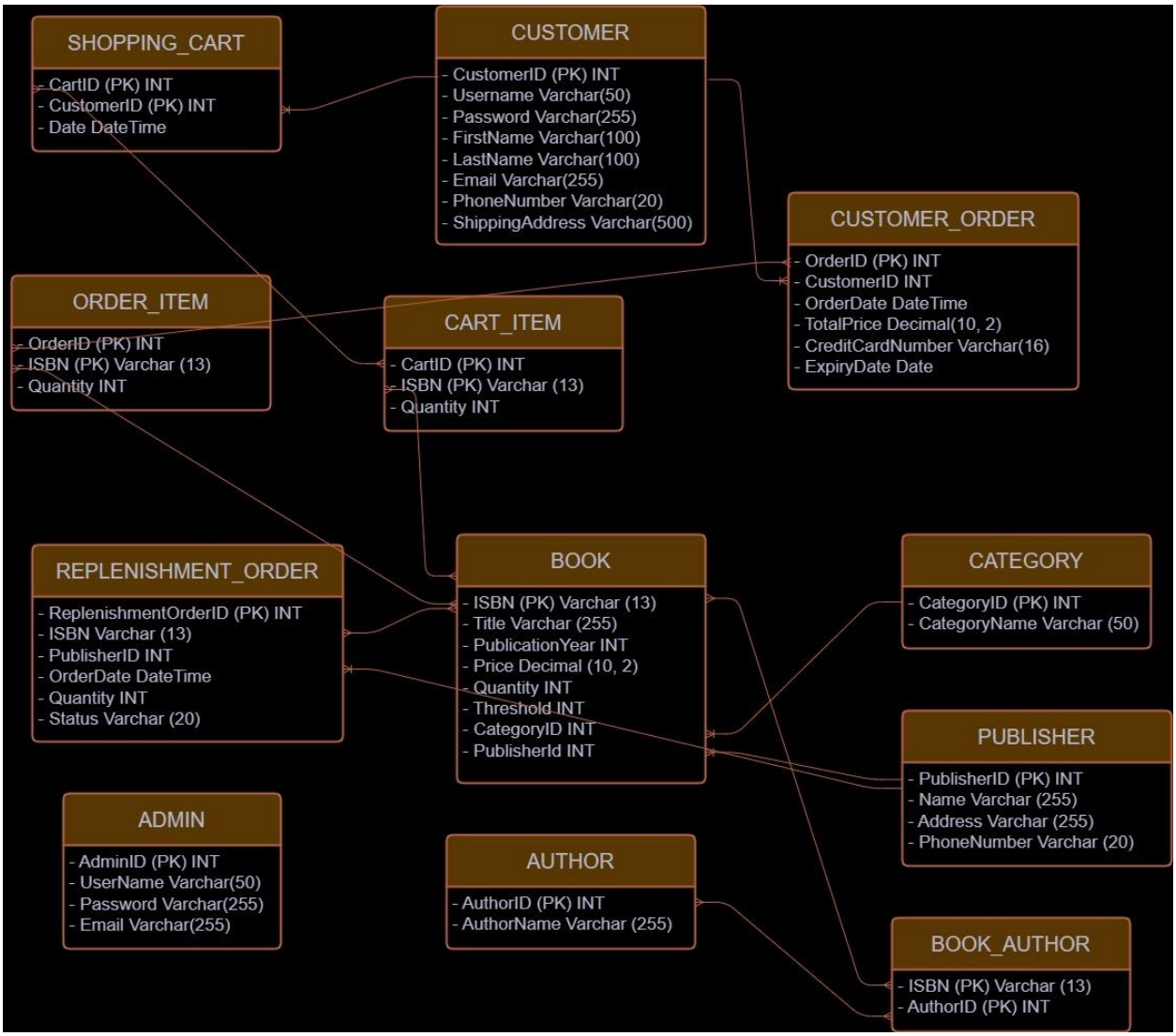
ER Diagram:


ISBN
Title
PublicationYear
Threshold
PubID
Name
Address
Needed
Price
BOOK
Publishes
PUBLISHER
Number
Supplies
Quantity
Category
Writes
AUTHOR
AuthorID
AuthorName
CATEGORY
CatID
CatName
Manages
Places
REPLENISHMENT_ORDER
ADMIN
AdminID
Email
Username
Password
Contains
Quantity
OrderID
Date
Status
CustomerID
CUSTOMER
Owns
SHOPPING_CART
CartID
CartDate
Phonenumber
Firstname
Lastname
Email
ShippingAddress
Places
ORDER
Sold
Quantity
CreditCardNumber
Expiry
OrderID
Date
Status
TotalPrice

ER Schema Mapping (Relational Schema):



Data Definition Language (DDL):

```
CREATE TABLE CUSTOMER (CustomerID INT AUTO_INCREMENT PRIMARY KEY, Username VARCHAR(50) NOT NULL UNIQUE, Password VARCHAR(255) NOT NULL, FirstName VARCHAR(100), LastName VARCHAR(100), Email VARCHAR(255) UNIQUE, PhoneNumber VARCHAR(20), ShippingAddress VARCHAR(500))
```

```
CREATE TABLE ADMIN (AdminID INT AUTO_INCREMENT PRIMARY KEY, Username VARCHAR(50) NOT NULL UNIQUE, Password VARCHAR(255) NOT NULL, Email VARCHAR(255) UNIQUE)
```

```
CREATE TABLE CATEGORY (CategoryID INT AUTO_INCREMENT PRIMARY KEY, CategoryName ENUM('Science', 'Art', 'Religion', 'History', 'Geography') NOT NULL)
```

```
CREATE TABLE PUBLISHER (PublisherID INT AUTO_INCREMENT PRIMARY KEY, Name VARCHAR(255) NOT NULL, Address VARCHAR(255), PhoneNumber VARCHAR(20))
```

```
CREATE TABLE AUTHOR (AuthorID INT AUTO_INCREMENT PRIMARY KEY, AuthorName VARCHAR(255) NOT NULL)
```

```
CREATE TABLE BOOK (ISBN VARCHAR(13) PRIMARY KEY, Title VARCHAR(255) NOT NULL, PublicationYear INT, Price DECIMAL(10, 2), Quantity INT DEFAULT 0, Threshold INT DEFAULT 5, CategoryID INT, PublisherID INT, FOREIGN KEY (CategoryID) REFERENCES CATEGORY(CategoryID), FOREIGN KEY (PublisherID) REFERENCES PUBLISHER(PublisherID))
```

```
CREATE TABLE SHOPPING_CART (CartID INT AUTO_INCREMENT PRIMARY KEY, CustomerID INT, Date DATETIME DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE)
```

```
CREATE TABLE CUSTOMER_ORDER (OrderID INT AUTO_INCREMENT PRIMARY KEY, CustomerID INT, OrderDate DATETIME DEFAULT CURRENT_TIMESTAMP, TotalPrice DECIMAL(10, 2), CreditCardNumber VARCHAR(16), ExpiryDate DATE, FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID))
```

```
CREATE TABLE CART_ITEM (CartID INT, ISBN VARCHAR(13), Quantity INT, PRIMARY KEY (CartID, ISBN), FOREIGN KEY (CartID) REFERENCES SHOPPING_CART(CartID) ON DELETE CASCADE, FOREIGN KEY (ISBN) REFERENCES BOOK(ISBN))
```

```
CREATE TABLE ORDER_ITEM (OrderID INT, ISBN VARCHAR(13), Quantity INT, PRIMARY KEY (OrderID, ISBN), FOREIGN KEY (OrderID) REFERENCES CUSTOMER_ORDER(OrderID), FOREIGN KEY (ISBN) REFERENCES BOOK(ISBN))
```

```
CREATE TABLE BOOK_AUTHOR (ISBN VARCHAR(13), AuthorID INT, PRIMARY KEY (ISBN, AuthorID), FOREIGN KEY (ISBN) REFERENCES BOOK(ISBN), FOREIGN KEY (AuthorID) REFERENCES AUTHOR(AuthorID))
```

```
CREATE TABLE REPLENISHMENT_ORDER (ReplenishmentOrderID INT AUTO_INCREMENT PRIMARY KEY, ISBN VARCHAR(13), PublisherID INT, OrderDate DATETIME DEFAULT CURRENT_TIMESTAMP, Quantity INT, Status ENUM('Pending', 'Confirmed') DEFAULT 'Pending', FOREIGN KEY (ISBN) REFERENCES BOOK(ISBN), FOREIGN KEY (PublisherID) REFERENCES PUBLISHER(PublisherID))
```

```
CREATE TRIGGER before_book_update BEFORE UPDATE ON BOOK FOR EACH ROW
BEGIN IF NEW.Quantity < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT =
'Update failed: Stock cannot be negative';
```

```
CREATE TRIGGER after_book_update AFTER UPDATE ON BOOK FOR EACH ROW
BEGIN IF OLD.Quantity >= NEW.Threshold AND NEW.Quantity < NEW.Threshold THEN
INSERT INTO REPLENISHMENT_ORDER (ISBN, PublisherID, Quantity, Status) VALUES
(NEW.ISBN, NEW.PublisherID, 50, 'Pending')
```

Features Implemented:

Admins:

1. Add new books.
2. Modify existing books.
3. Place orders on books when quantity decreases below threshold.
4. Confirm orders.
5. Search for books.
6. Generate system reports.

Customer:

1. Edit personal information
2. Search for books through attributes
3. Manage his shopping cart.
    a. Add books to cart.
    b. View items in cart.

   c. View individual and total items in cart.

   d. Remove items from cart.

4. Logout from the system.

5. Check out.

6. View past orders.